

Full Progress Report: FRB Host Galaxy Analysis

GitHub Copilot

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1 Photutils Inclination Analysis

Comparison between photutils, GALFIT, and SDSS inclinations.

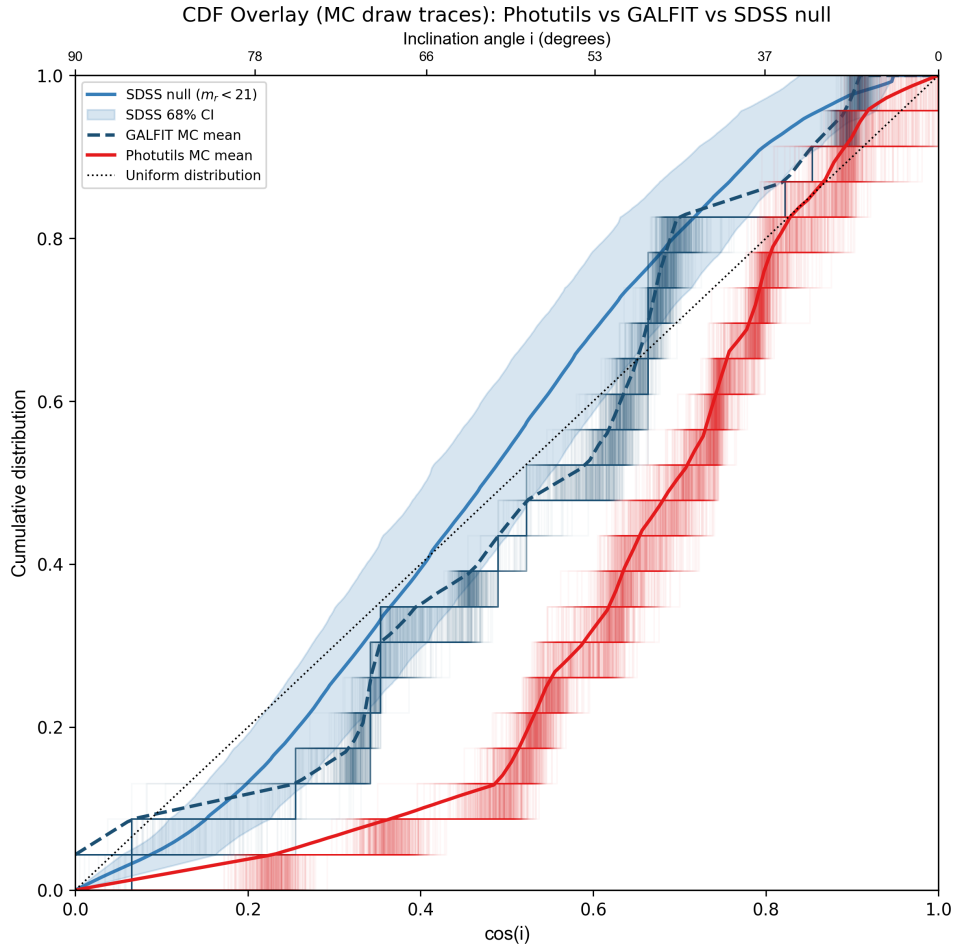


Figure 1: GALFIT and Photutils inclination comparisons against SDSS, including Monte Carlo uncertainty tracks.

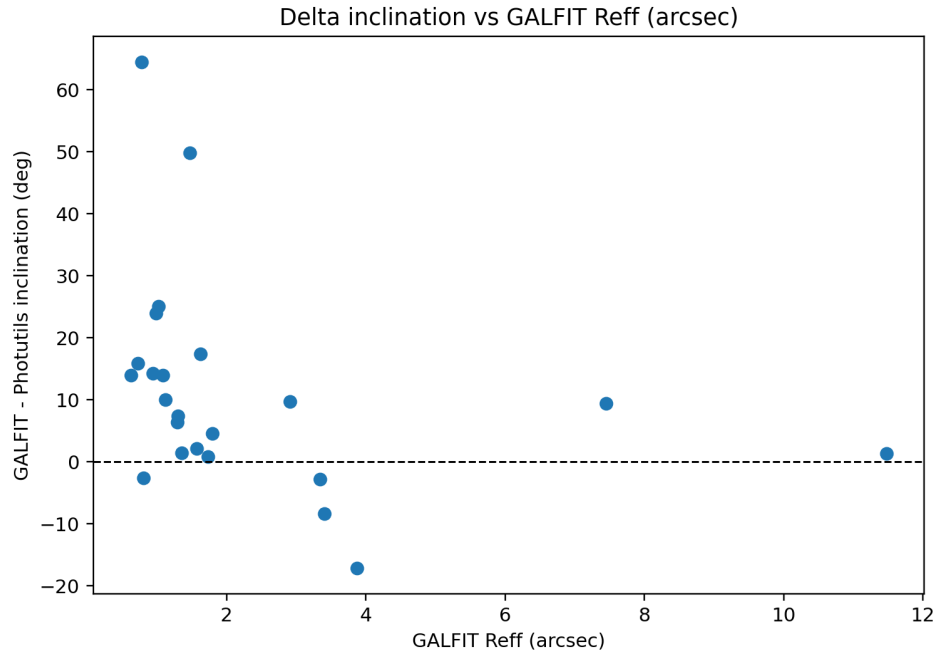


Figure 2: Difference in inclination (GALFIT - Photutils) vs. GALFIT effective radius in arcseconds.

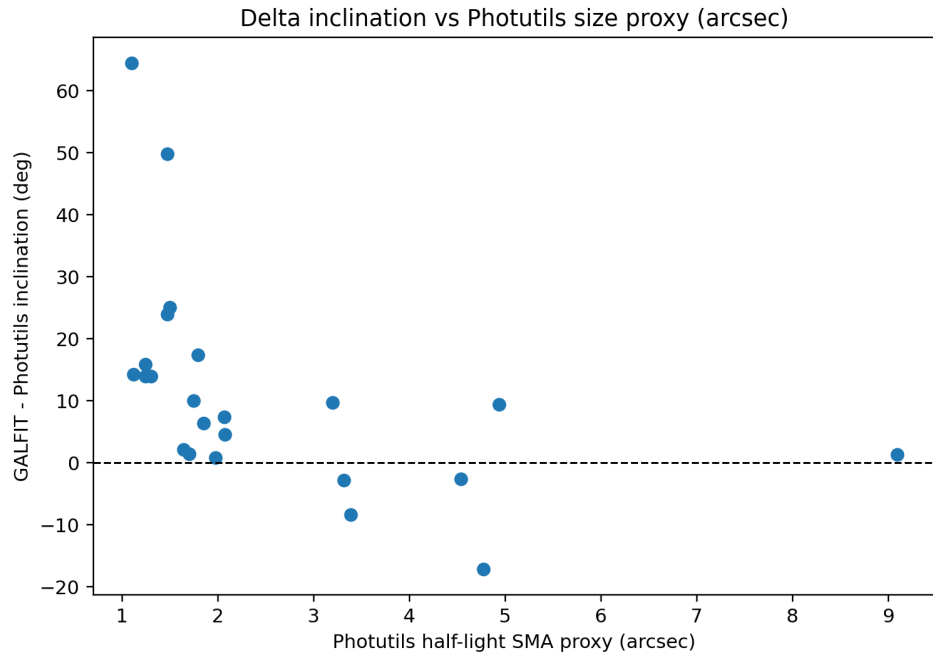


Figure 3: Difference in inclination (GALFIT - Photutils) vs. Photutils semi-major axis proxy in arcseconds.

2 Legacy Survey vs. GALFIT Comparison

Even as n changes massively, R_{eff} remains nearly the same in Legacy Survey fits; with model and residual images looking so similar, the fitting behavior is confusing.

FRB	n_{survey}	$R_{\text{eff,survey}}$ (arcsec)	$R_{\text{eff,galfit}}$ (arcsec)	i_{survey} (deg)	i_{galfit} (deg)
20180924B	3.3241782188415527	0.7392469048500061	0.6288	48.12091274655155	33.63
20190102C	0.5	0.9823504686355592	0.98774	64.70865680540318	80.24
20190608B	3.487734794616699	1.5951251983642578	1.29428	54.32229704490156	32.52
20190714A	0.5335768461227417	1.0340769290924072	1.0873000000000002	68.70084498763808	70.75
20191001A	0.5	1.1012290716171265	1.73182	82.05605101232584	46.79
20200906A	0.745732307434082	1.539622187614441	1.62702	74.90203346851389	77.62
20210320C	1.0	0.7692122459411621	0.94582	0.0	45.1
20210410D	1.0	0.7306411266326904	0.7833800000000001	74.99381150093153	90.0
20211203C	1.0	0.6896117925643921	0.72312	62.89032117370526	56.24
20211212A	1.1396939754486084	3.211979866027832	3.40862	35.08222954155959	32.52
20220310F	1.0	0.9910320043563844	1.1266	49.871877685030114	70.02
20220509G	2.466068983078003	4.925990104675293	2.91344	59.34975965734881	62.11
20220914A	2.032114028930664	1.0738459825515747	1.02704	63.08149375226626	59.94
20220920A	1.0	1.6488510370254517	1.79994	0.0	31.39
20221012A	2.2670769691467285	1.2647240161895752	1.30476	59.52912541944941	59.21

Table 1: FRBs with survey matches only (match_found=True).

Legacy tends to fit very low-SNR regions, which can break up galaxies (as seen in the largest-deviation cases); GALFIT still provides the better fit overall.

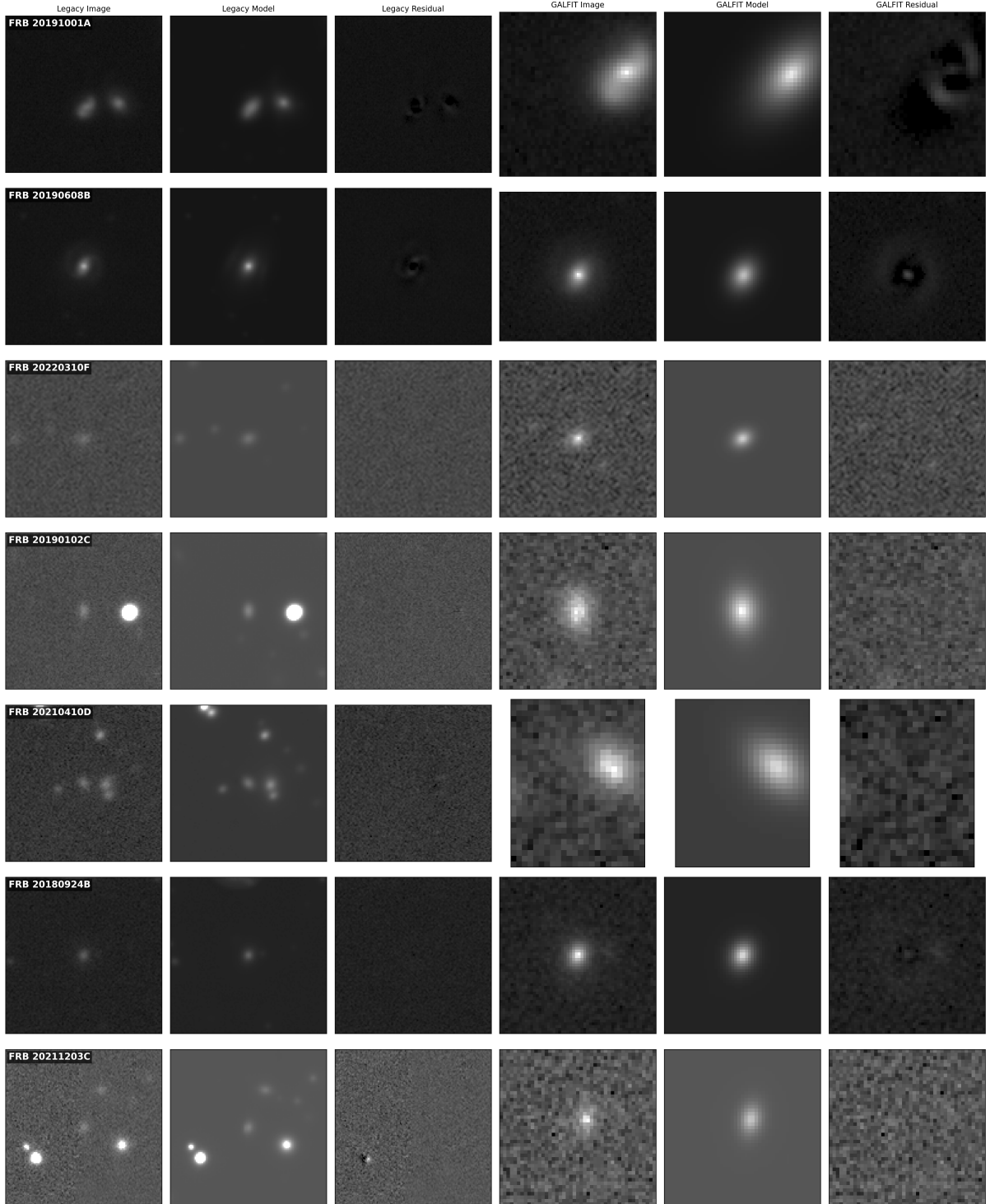


Figure 4: Top 7 non-REX Legacy-vs-GALFIT IMR diagnostics, with FRB names shown on each row.

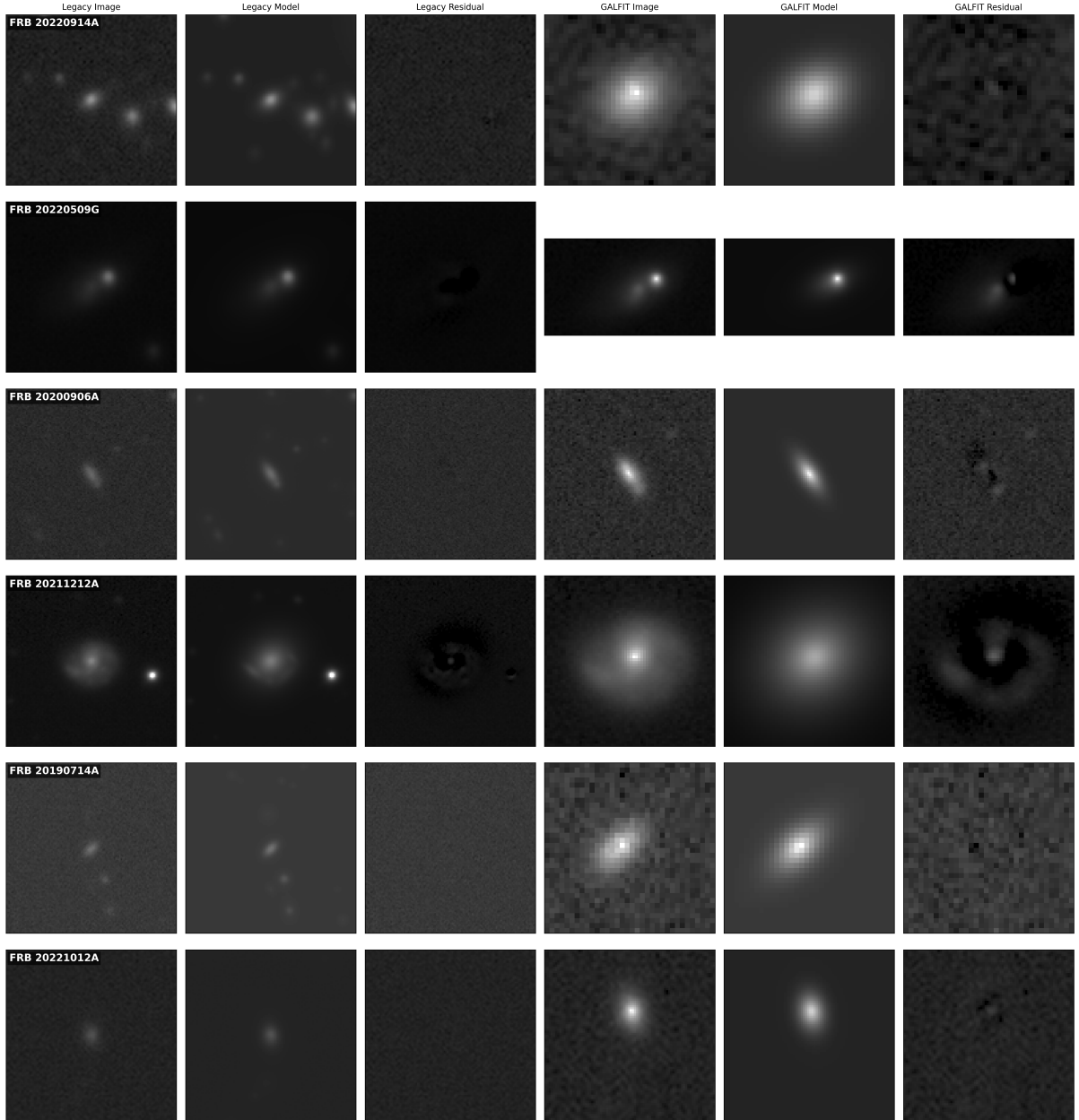


Figure 5: Remaining 6 non-REX Legacy-vs-GALFIT IMR diagnostics, with FRB names shown on each row.